

Amar Vashishth

Senior Software Engineer

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Profile

Senior backend and platform engineer with 7+ years building high-throughput commerce systems, messaging infrastructure, Kubernetes services, and internal GenAI engineering tools. Experienced in zero-downtime migrations, production reliability, secure data workflows, and LLM-powered automation. Python, Java, Go, C++, Springboot, Kafka, K8s, MySQL, Couchbase

Education

Master of Technology | Computer Science and Engineering

2017 – 2019

Indian Institute of Technology Madras

Chennai, Tamil Nadu

- Graduate coursework and research focus on deep learning, pattern recognition, probabilistic methods, Bayesian inference, Markov chains, and linear algebra for ML systems.
- Earned admission through the highly selective IIT entrance process, ranking in the top 0.1 percentile among 500K+ national candidates across 14 computer science subjects.

Work Experience

Applications Engineer

July 2023 – Present

Rakuten

Tokyo, Japan

– Generative AI Engineering Tools

Technologies: OpenAI, Claude, Gemini, Qwen, DeepSeek, Python, Chrome Extension

- * **Real-time production log and metrics analysis with AI**, converting noisy operational signals into structured engineering reports.
- * Built **Chrome extension to integrate LLM assistance** into daily engineering workflows for contextual checks, documentation tasks, and pull-request review support.

– High-Scale Commerce Platform Modernization

Technologies: Java, Spring Boot, Python, Kafka, React, Oracle, Couchbase, Redis

- * Modernized browsing-history and gift-history systems, each with several services, supporting **34,000 requests per second** for Rakuten's commerce storefront.
- * Executed a **zero-downtime, rollback-safe Couchbase-to-Oracle migration** while preserving continuity for high-traffic user-history flows.
- * Migrated *ashiato.rakuten.co.jp* storefront capabilities to React based frontend.

– Cloud Migration, Delivery, and Reliability

Technologies: Kubernetes, Jenkins, AWS, Istio, Envoy, GSLB, Apache

- * Migrated critical commerce services from **bare-metal infrastructure to Cloud-infra with Kubernetes** orchestration, improving deployment consistency and operational scalability.
- * Served as acting **Tech Lead** coordinating technical priorities, incident response, onboarding, and service continuity.
- * Improved **Super Sale readiness** through observability enhancements, security hardening, vulnerability remediation, and large-scale load testing.

Senior Software Engineer

July 2019 – July 2023

Myntra (Walmart)

Bengaluru, India

– Messaging, Streaming, and Data Security Platforms

Technologies: Java, Python, Go, JavaScript, Kafka, Security, React

- * Architected and maintained enterprise messaging infrastructure handling **60,000 requests per second** for business-critical workloads.
- * Built and maintained **SDKs** in Java, Python, JavaScript, and Go to standardize messaging system adoption.
- * Designed and implemented self-service cross-datacenter Kafka topic replication and secure handling for PII and sensitive data workloads.

- Business-Critical Backend Systems

Technologies: Java, Spring Boot, MySQL, REST API Security, C#

- * Developed PLM backend services processing 100,000 requests per minute, **replacing PRM360 workflows**.
- * Built KnowYourClient for real-time API protection and client metadata management across **250K monthly requests** without service modifications.
- * Delivered a **C# Excel automation plugin** that reduced manual validation and upload effort for PLM business users.

- Reliability, Observability, and Platform Operations

Technologies: ELK Stack, Prometheus, Grafana, Docker, MySQL 8, Load Testing

- * Integrated ELK logging, StatsD, InfluxDB, Prometheus, and Grafana with email and Slack alerting for mission-critical services.
- * Containerized backend services, **led MySQL 5.7 to MySQL 8 migration** work, and performed **Kafka brokers stress tests** before major sale events.

Projects and Research

Deep Learning Thesis: Nutritional Analysis of Indian Cuisines

Technologies: CNNs, Computer Vision, TensorFlow, Data Engineering, Rule-Based Reasoning

- Built a CNN workflow to identify Indian dishes from images and support automated nutritional analysis.
- Estimated calories by combining model outputs with deterministic nutrition rules and rule-based reasoning.
- Created Indian-dish datasets through government-agency coordination, web crawling, and label cleanup.

Advanced Pattern Recognition & Deep Learning Academic Projects

Technologies: Python, TensorFlow, PyTorch, NumPy, SciPy, Scikit-Learn

- Designed and evaluated neural architectures including Sequence-to-Sequence (Seq2Seq) Networks, text embeddings with word2vec, and image classification via Fashion MNIST using CNNs.
- Built and analyzed probabilistic architectures including Bayesian Neural Networks, Restricted Boltzmann Machines (RBMs), Hidden Markov Models (HMMs), and Gaussian Mixture Models (GMMs) for advanced pattern identification tasks.

Probabilistic ML and Robotics Academic Projects

Technologies: Bayesian Inference, Markov Chains, SPARQL, Java, Gmail API

- Implemented SPARQL/RDF query systems and a Java-based Bayesian email classifier integrated with the Gmail API.
- Engineered autonomous robot navigation systems utilizing real-time obstacle avoidance rules.

Skills

Languages: Java, Python, Go, JavaScript, TypeScript, C#, C/C++, Rust

Backend and Web: Spring Boot, FastAPI, React, Node.js, REST APIs, Nginx, Apache HTTP Server

AI and Machine Learning: LLM application design, AI-assisted code analysis, OpenAI, Claude, Gemini, Qwen, DeepSeek, CNNs, Bayesian inference, Markov chains, Matlab

Data and Messaging: Kafka, RabbitMQ, Redis, MySQL, Oracle, MongoDB, Couchbase

Cloud and DevOps: Kubernetes, Docker, Jenkins, GitHub Actions, Ansible, AWS, Azure

Reliability and Testing: Prometheus, Grafana, ELK Stack, StatsD, InfluxDB, Gatling, k6, JUnit, Mockito